Reversing a conditional probability

High school · probability · Synthesis problem · 07

Given information. Choose A with probability $\frac{1}{3}$ and B with probability $\frac{2}{3}$. Urn A contains 3 red balls and 1 blue; B contains 1 red and 3 blue. After observing one red ball, the probability that A was chosen is $\frac{3}{5}$.

Problem · Investigate, justify, conclude. Use the two urns from the example. Choose an urn once, then draw twice with replacement from that same urn. Find the probability of two red balls, then the probability that A was chosen given two red balls. Compare with observing one red ball and justify the products used.